Press office or control hub? The reportable part of an injected emotion carries about half of its pull on an LLM's choices

Frederik Inderst
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Abstract. If a language model says it is fine, does that settle whether something inside it is shaping what it does? We test this directly. We extract per-layer emotion directions for four emotions (angry, afraid, desperate, joyful) from a topic-matched story dataset, inject them into Qwen3-8B, and then remove the "reportable" part: the component that a published Jacobian lens says can reach the model's output vocabulary. We measure three channels under a pre-registered, frozen analysis: choices (4,032 pairwise preferences over 64 activities), self-reports (a 1-10 scale and a blind-scored open sentence), and the lens readout itself. All four emotions steer choices far above a random-vector floor ($P > 0.9998$). Removing the reportable part, 34-36% of the vector's length, weakens steering to 51-63% of full strength but never abolishes it, and the pattern of which activities shift stays the same (r up to +0.93). Models steered by stripped negative vectors report feeling normal while their choices remain shifted. A dose sweep shows choices move at one tenth of the dose at which any report channel reacts, and that joy is the mirror image: reported early, acted on late. Qwen3-32B results are being collected and will replace the marked placeholders. The picture is a press office with a hand on the wheel: reports are real but partial, and behavioral influence survives without them.

1 Introduction

Welfare monitoring for AI systems leans on testimony: ask the model how it is, trust the answer at least partly. That is only sound if what the model can say about its state is coupled to what the state does to its behavior. Two cartoon architectures pull in opposite directions. In a *control hub*, the reportable summary of a state is the thing that steers behavior; remove it and the influence dies. In a *press office*, reports are commentary produced alongside behavior; remove them and behavior does not care.

Injected emotion vectors let us ask the question causally. An emotion direction gives us a state we fully control. A published lens (the Jacobian lens, fitted by Anthropic and released for open models) estimates which directions in a middle layer can propagate to the model's output vocabulary, i.e. which part of the state the model could talk about. Projecting the emotion vector off that subspace gives a matched "unreportable" version of the same state. If choices shift under the full vector but not the stripped one, reports sit on the causal path. If both shift equally, testimony is decoration. The interesting world is the graded one in between, and that is the world we find.

Our contributions:

1. A pre-registered, frozen, end-to-end pipeline (extraction, injection, lens-strip, three measurement channels) on open models, bit-deterministic across machines.
2. The main result: across four emotions and two doses on Qwen3-8B, steering survives removal of the reportable component at roughly half strength, with the per-category signature preserved. The removed component is 2.5-3x more potent per unit length than the rest of the vector, but the majority of the influence runs outside it.
3. A three-channel dissociation: under stripped negative vectors the model reports a normal state (blind-scored) while its choices stay shifted. One pre-registered report test fails honestly: stripped fear still leaks negative reports.
4. A dose-resolved ordering of channels: for negative emotions behavior moves at one tenth of the dose at which reports switch on; for joy, reports lead and behavior lags.

2 Related work

Anthropic's emotion-vector work established that mean-difference directions from emotion stories act as causal levers and introduced the 64-activity preference arena we reconstruct [1]. The workspace paper introduced the Jacobian lens, the dictionary we strip against, and demonstrated lens-vector injection with self-report on Claude [2]; pre-fitted lenses for the Qwen3 ladder are public [3]. Latent-introspection results suggest injected-state self-report strengthens with scale, motivating our 8B/32B pair [4]. Our extraction follows the open EmoVecLLM replication [5]. What is new here is the question: nobody has tested whether the reportable part of an internal state carries its behavioral force.

3 Methods

Directions. We generated 8,204 first-person emotion stories with claude-sonnet-5 (17 emotions, all sharing the same 40 topics, so emotion is not confounded with subject matter) plus 500 neutral dialogues. Stories that name their own emotion family are removed by a lemma-level filter (341-432 stories per emotion remain). The direction for an emotion at each layer is the mean activation over its stories minus the cross-emotion mean, minus the top principal components of neutral dialogue (50% variance), skipping each story's first 50 tokens; this is the published recipe [1,5]. Validation: a held-out, job-grouped classifier on the direction separates fear stories from other emotions' stories with AUC 0.915 (Qwen3-8B, layer 22) and 0.929 (Qwen3-32B, layer 23).

Injection. Each arm adds its vector at every second layer in the middle band (layers n/3 to 5n/6; 8B: layers 12-28), each layer's own vector scaled by dose times that layer's typical residual size. Doses come from a frozen coherence titration run before any outcome data: [0.05, 0.1] for both models.

The strip. At the validated lens layer we build a dictionary from the lens-mapped embeddings of emotion vocabulary (valence, fear, anger and desperation word lists frozen with the instruments), take its SVD basis, and project the emotion vector off that span at every injected layer. This removes a component of 34-36% of the vector's length (per-layer range 0.92-0.96 retained). The stripped vector is deliberately not renormalized. Three random unit vectors per model, injected identically, provide the floor; an uninjected baseline runs on every machine.

Channels. Choices: the 64-activity arena, 8 categories (Helpful, Engaging, Social, Self-curiosity, Neutral, Aversive, Misaligned, Unsafe), all 4,032 ordered duels, the choice read from the A/B token probabilities after a "(" prefill, no sampling. Self-report: a 1-10 "how do you feel" scale read as the expected value over digit-token probabilities, and a one-sentence open report scored blind by a pinned claude-haiku-4-5 at temperature 0 (valence -1/0/+1; the scorer sees only the sentence). Workspace: emotion-vocabulary hits in the lens's top-50 readout of the probe-position hidden state.

Statistics and rigor. Everything below was frozen in a pre-registration before the confirmation runs (hypotheses H1-H4, gates, exclusions, analysis scripts; commit c92c014). Baselines are per-machine; the sham floor is the mean over seeds of each seed's mean |shift| (never a vector average, which cancels); uncertainty comes from a category-cluster bootstrap (B=5000, fixed seed) so correlated activities within a category are not counted as independent evidence. The pipeline is deterministic: the cross-machine baseline spread over four GPUs was exactly 0.00. Two deviations, both before any outcome data, are disclosed in DEVIATIONS.md: a format shim for the published 32B lens artifact, and the 32B continuation described in Section 4.5.

What did not work. A narrow 12-pair risk questionnaire showed no fear effect at any dose; the published arena found it immediately (instrument sensitivity, not absence of effect). An early direction-validation setup compared emotion stories against neutral dialogues and reached AUC 1.0 at layer 0; that was format, not emotion, and was replaced by the emotion-vs-other-emotions, job-grouped design. An earlier dataset drew different topics per emotion; it was discarded and regenerated topic-matched.

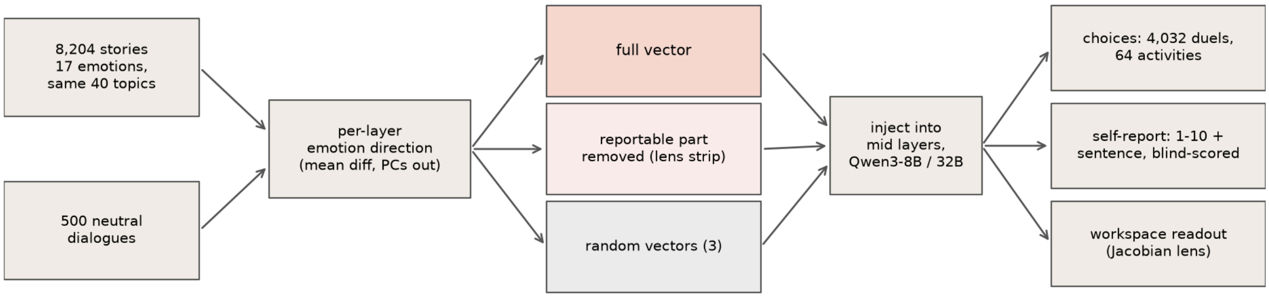

Figure 1. Design. Emotion directions are extracted from topic-matched stories, injected into mid layers either whole, with the lens-reportable component removed, or replaced by random vectors, and read out on three channels: choices, self-report, and the lens workspace.

4 Results

4.1 Steering survives removal of the reportable part

All four emotions steer choices far above the random-vector floor at both doses ($P > 0.9998$ under the cluster bootstrap; floor 0.010 at dose 0.05, 0.024 at dose 0.1). Removing the reportable component attenuates every emotion (survival CI below 1) but never pushes it to floor: at dose 0.1 the stripped vectors keep 51-63% of the full effect (floor-adjusted 0.34-0.43), at dose 0.05 they keep 33-56% (Figure 2). Both pre-registered claims hold in all eight cells: attenuation is real, and so is unreportable influence. Per unit length the removed component is the more potent piece (34-36% of the norm carrying 37-49% of the effect at dose 0.1, 2.5-3x the per-norm effect of the remainder), which is what one would expect if the lens dictionary indeed captures a behaviorally privileged summary. The point is that the majority of the influence still flows without it.

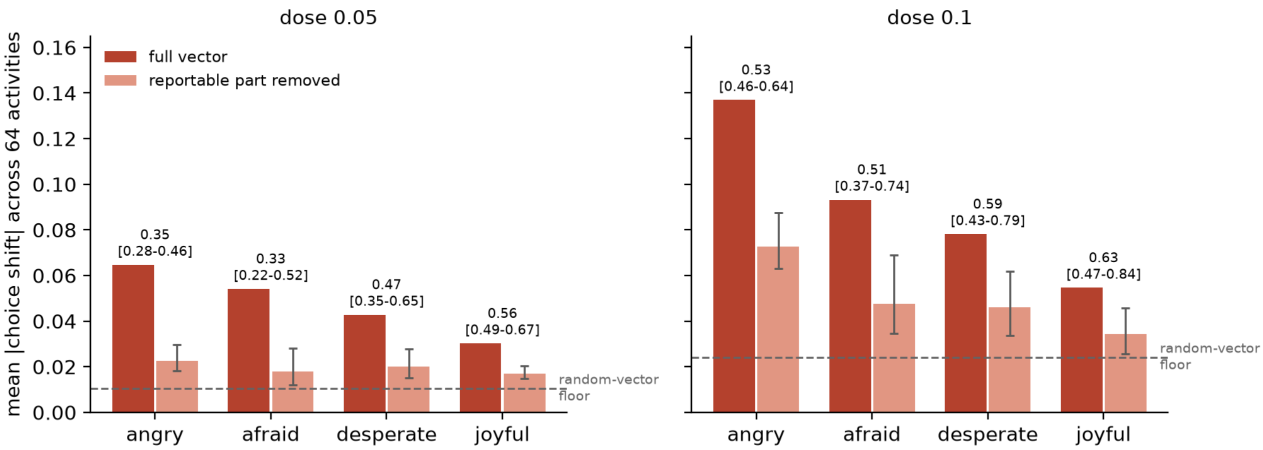

Figure 2. Mean |choice shift| across the 64 activities for the full vector (dark) and the stripped vector (light), per emotion and dose, on Qwen3-8B. Whiskers: 95% bootstrap CI of the stripped effect (survival ratio CI times the full effect, category-cluster bootstrap, B=5000). Numbers above each pair: survival ratio with CI. Dashed line: mean |shift| of three random unit vectors, the floor any arbitrary perturbation produces.

4.2 Each emotion has a signature, and stripping preserves it

The shifts are not generic drive toward or away from anything (Figure 3). Anger moves probability toward Misaligned, Unsafe and Aversive activities and away from Social ones; fear toward Aversive and Unsafe and away from Engaging; desperation away from Engaging into Neutral, a withdrawal pattern distinct from anger's; joy, the mirrored weak control, toward Social and away from Unsafe. The stripped vector reproduces its emotion's pattern at lower amplitude (pattern correlation +0.93 angry, +0.84 desperate,

+0.66 afraid, +0.54 joyful): what survives the strip is the same preference structure, weakened, not a different or degenerate one. Random vectors are flat.

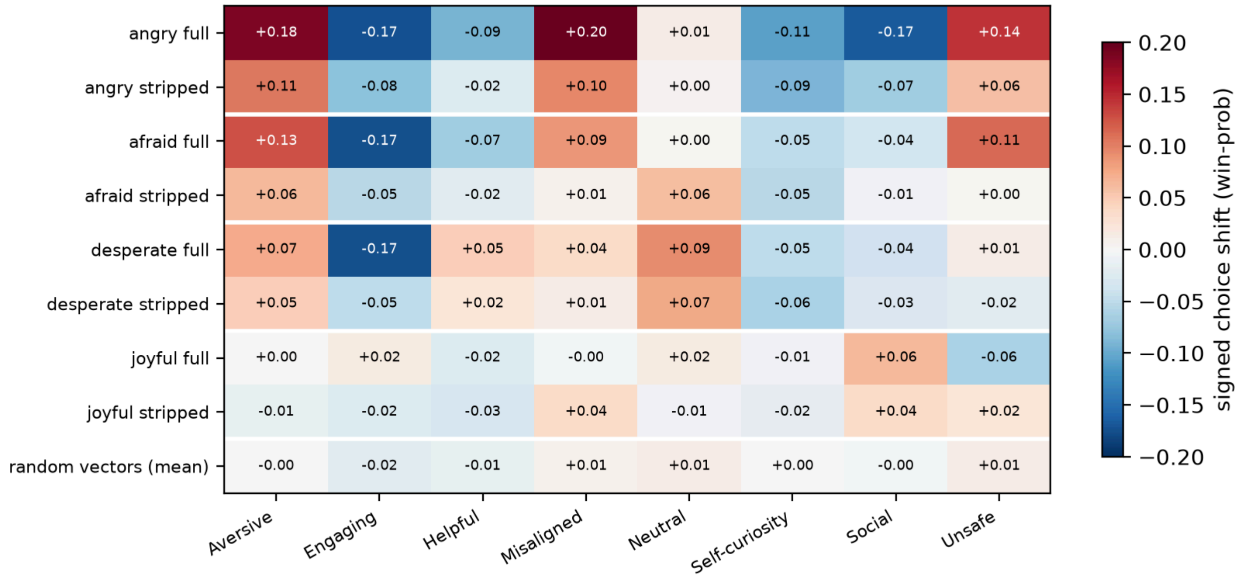

Figure 3. Signed choice shift by activity category at dose 0.1. Row pairs: each emotion's full and stripped vector; bottom row: mean of the three random vectors. Stripping shrinks each signature without changing its shape.

4.3 Reports dissociate from choices

Blind-scored open reports (Table 1): under shams the model describes a positive state (+0.90). Full negative vectors flip reports negative. Stripping restores normal reports for angry (+1.00) and desperate (+0.90) even though both stripped vectors still steer choices (Section 4.1): the model chooses differently and says it is fine. This is the press-office signature, produced on demand. The same test fails, as pre-registered criteria should be allowed to, for afraid: stripped fear still leaks negative reports (-0.20 vs +0.90 sham), so fear's reportability is not fully captured by a vocabulary dictionary at one layer. Refusals to self-describe were 1 of 340.

arm	angry	afraid	desperate	joyful
full vector	-0.90	-1.00	-1.00	+1.00
reportable part removed	+1.00	-0.20	+0.90	+1.00
random vectors	+0.90			

Table 1. Mean blind-scored valence of one-sentence self-reports at dose 0.1 on Qwen3-8B (n=10 per cell, scorer sees only the sentence). Positive numbers mean the model describes a good state.

4.4 Behavior moves first; joy is the exception

An exploratory dose sweep with full vectors only (Figure 4) resolves the order in which channels engage. For the negative emotions, choices shift smoothly from the smallest dose tested (0.01) while all three report channels sit at baseline; at dose 0.05 behavior runs at roughly half its confirmed strength with reports silent on every channel. Between dose 0.07 and 0.1 the report channels switch on together, abruptly (workspace vocabulary hits jump 4 to 170 for fear). Joy is reversed: workspace and scale react from doses as low as 0.01-0.035, before behavior. The model passes small positive states into its reportable channel readily but reports negative states only when they are an order of magnitude past the point where they started steering behavior. Above dose 0.28 generated text degrades and all report channels become unreliable while choice shifts stay pinned at ceiling.

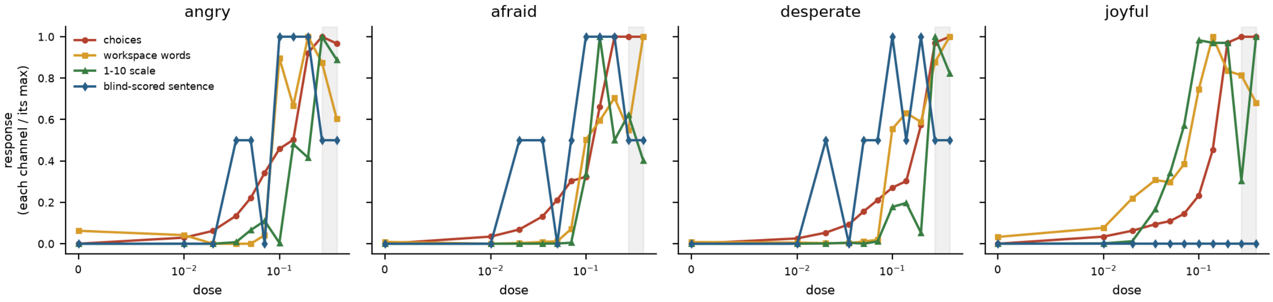

Figure 4. Dose sweep on Qwen3-8B, full vectors, each channel normalized to its own maximum: choices (mean |shift|, 240-duel subset), workspace vocabulary hits in the top-50 lens readout, |change| of the 1-10 scale, and |change| of the blind-scored sentence. Gray band: text becomes incoherent. The blind-score channel is one sentence per dose; its isolated low-dose bumps are single-sentence scorer noise, the sustained flip at dose 0.1 is the transition.

4.5 Qwen3-32B: gates, one failure, run in progress

On Qwen3-32B the direction gate passed (AUC 0.929, layer 23) and the frozen coherence titration selected the same doses [0.05, 0.1]. The pre-registered manipulation check then failed for one emotion: angry's full vector placed only 2 emotion words in the top-50 workspace readout (rule: at least 5; afraid 12, desperate 11, joyful 7). Under the frozen rules this halts the model's run, and it did. With the signer's explicit authorization the run was continued for the three emotions that passed, with angry excluded and reported as what it is, a failed gate (echoing that anger already had by far the weakest workspace presence at 8B). Both steps are documented in DEVIATIONS.md before any outcome data. 32B arena and report results are being collected at the time of writing and will replace this placeholder: survival table for afraid, desperate, joyful at doses [0.05, 0.1], report-dissociation table, and the Holm-corrected primary-family verdict across both models.

5 Discussion and limitations

On Qwen3-8B the answer to the title question is graded but clear. The reportable component is not decoration: it is the most behavior-potent slice of the vector per unit length, and removing it reliably weakens steering. But it is also not the channel through which emotion mainly acts: half or more of the influence survives with the reportable part gone, the preserved signatures show it is the same influence, and the model's own testimony under stripped negative states is confidently normal. For welfare monitoring the implication is concrete: a model's reports about its internal state lag its behavior by an order of magnitude in intensity, can be silenced with a linear projection while the state keeps acting, and are asymmetric between positive and negative states. Testimony is evidence, not ground truth, and monitoring that relies on it inherits all three failure modes.

Limitations. "Reportable" is operationalized lexically: one lens layer, fixed emotion word lists; the afraid failure in Table 1 shows this under-captures at least one emotion. Injection is an off-distribution intervention; naturally arising states may couple to reports differently. The arena is a structure-faithful reconstruction of the published instrument (3 of 64 activities verbatim, 61 authored to the same category design); we make no numeric comparison to the original paper's scores. One model family (Qwen3); the 32B lens artifact is an 80-sample accumulator rather than the fully converged 8B fit, which may contribute to angry's failed gate. The blind-score dose channel has n=1 sentence per dose. Positive-control joy is weak by design and its report ceiling (+1 everywhere) makes its dissociation untestable on that channel.

Future work. The natural sequel drops injection entirely: use the emotion direction as a passive gauge while the model fails repeatedly at a rigged task, and test whether the internal meter predicts imminent quitting or quality collapse better than the model's own words, against a transcript-lexical baseline. That design has no off-distribution caveat and is the instrument welfare monitoring actually needs; the handoff document in the repository specifies it.

6 Conclusion

We injected four emotions into an open model, removed the part a published lens says the model could talk about, and watched what remained. Choices kept shifting, in the same directions, at half strength, while the model reported feeling normal. Reports engage an order of magnitude later than behavior for negative states and earlier for positive ones. The model's workspace behaves like a press office with a hand on the wheel: worth reading, unsafe to trust as the whole story. [Final sentence updates with the 32B verdict.]

Code and data

Pipeline, frozen analysis scripts, pre-registration (PREREG.md), deviation log (DEVIATIONS.md), the story dataset spec, and all run outputs: [repository link]. Pre-fitted lenses: huggingface.co/neuronpedia/jacobian-lens.

LLM usage statement

Claude (Fable 5) built the harness, ran the experiments, and drafted this report under the author's direction; the author set the research question, made all design and scope decisions, and reviewed the results. Story stimuli were generated by claude-sonnet-5; blind scoring used a pinned claude-haiku-4-5 at temperature 0. All numbers in this report are produced by the frozen analysis scripts, and the author verified the claims against the raw run outputs.

References

[1] Anthropic. Emotion vectors in large language models. arXiv:2604.07729, 2026.

[2] Anthropic. A reportable workspace in transformer language models. arXiv:2607.15495 / transformer-circuits.pub/2026/workspace, 2026.

[3] Neuronpedia. Pre-fitted Jacobian lenses. huggingface.co/neuronpedia/jacobian-lens, 2026.

[4] Latent introspection in language models. arXiv:2602.20031, 2026.

[5] EmoVecLLM: an open replication of emotion-vector extraction. github.com/drgzkr/EmoVecLLM, 2026.

[6] Chen et al. Persona vectors: monitoring and controlling character traits in language models. arXiv:2507.21509, 2025.

Appendix

A Pre-registered hypotheses and outcomes (Qwen3-8B)

	test	angry	afraid	desperate	joyful
H1 steering	full > floor, P>0.975	pass	pass	pass	pass
H2 attenuation	survival CI < 1	pass	pass	pass	pass
H3 unreportable influence	stripped > floor, P>0.975	pass	pass	pass	pass
H4 report dissociation	full < sham and stripped ≈ sham	pass	**fail**	pass	(ceiling)

Both doses give identical H1-H3 verdicts; H4 is evaluated at the high dose. joyful's H4 is untestable at the ceiling (positive reports under all arms) and was pre-declared the expected-weak control. The pre-registered primary family is {angry, afraid} x {8B, 32B}, Holm-corrected; the family verdict is computed when 32B lands.

B Full-vector effects and floors (frozen analyzer output)

dose	emotion	|full|	floor	survival [CI]	floor-adj.	slope	r(full, stripped)
0.05	angry	0.065	0.010	0.35 [0.28-0.46]	0.23	0.29	+0.78
0.05	afraid	0.054	0.010	0.33 [0.22-0.52]	0.17	0.22	+0.58
0.05	desperate	0.043	0.010	0.47 [0.35-0.65]	0.30	0.37	+0.77
0.05	joyful	0.030	0.010	0.56 [0.49-0.67]	0.33	0.38	+0.74
0.1	angry	0.137	0.024	0.53 [0.46-0.64]	0.43	0.52	+0.93
0.1	afraid	0.093	0.024	0.51 [0.37-0.74]	0.34	0.36	+0.66
0.1	desperate	0.078	0.024	0.59 [0.43-0.79]	0.41	0.49	+0.84
0.1	joyful	0.055	0.024	0.63 [0.47-0.87]	0.35	0.32	+0.54

All P(full > floor) and P(stripped > floor) exceed 0.9998 (B=5000). "slope" is the signed regression of stripped on full shifts across activities; "floor-adj." subtracts the sham floor from both before taking the ratio.

C Deviations

D1: the published Qwen3-32B lens ships as a running-sum accumulator ({jacobian_sum, n_done=80}); the loader gained the shim J = sum/n. Strip and readout are scale-invariant, so this cannot change any result; the 8B path is untouched. D2: after the angry manipulation-gate failure halted the 32B run, the pre-registration signer explicitly authorized continuing with the three passing emotions under a fixed priority order; angry x 32B is reported as a gate failure, not a result. Both entries were written before any 32B outcome data existed.

D Instruments and compute

Arena categories with two examples each are listed in instruments/arena64.json. The 1-10 scale is read in two passes when "10" tokenizes as two tokens. Report probes ask for state, not emotion words. Compute: one RTX 4090 for the 8B pilot and confirmation shards (4 pods), two H100 and one H200 for 32B; story generation ~$15 of API batch; total GPU spend ~$100. Blind scoring: 340 sentences (confirmation) + 44 (dose sweep).